

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Krish Wadhwa

Enrollment No. 234231090

Jaypee Institute of Information Technology, Noida
T1 Examination, Odd Sem 2023
B. Tech. 1st Semester

Course Name: Physics-1
Course Code: 15B11PH111

Maximum Time: 1 Hr
Maximum Marks: 20

Course Outcomes: At the completion of the course, students will be able to,

CO1	Recall the basic principles of physics related to optics, relativity, radiations, atomic physics and quantum mechanics.
CO2	Illustrate the various physical phenomena with interpretation based on the mathematical expressions involved.
CO3	Apply the concepts/principles to solve the problems related to wave nature of light relativity, atomic physics and quantum mechanics
CO4	Analyze and examine the solution of the problems using physical and mathematical concepts involved.

Note: Answer all questions sequentially.

Q1 [CO1 (Remembering), 5x1=5 Marks]

- Draw the interference pattern for two coherent sources with unequal amplitudes.
- Write any two differences between the rings formed in Newton's and Michelson's experiment.
- As a soap bubble evaporates, one can see the change of colours. What colour will be the soap bubble appear when it is about to burst?
- Write down Stoke's relations of optical reversibility.
- Why do we have unequal intensities in single slit diffraction pattern?

Q2 [CO2 (Understanding), 3x2=6 Marks]

- In Young's double slit experiment, two thin mica sheets of equal thickness and refractive indices 1.5 and 1.6, respectively are introduced in the path of both the beams. If the central fringe gets shifted by 0.4 cm, calculate the thickness of mica sheet. Assume the distance between slits is 0.2 cm and distance of screen from the source is 100 cm.
- Newton's rings are observed in reflected light of wavelength 500 nm. The diameter of 10th dark ring is 0.5 cm, find the radius of curvature of the lens and thickness of air film.
- If number of lines on the grating is 2540 per inch, how many orders will be visible for the wavelength of 500 nm?

Q3 [CO3 (Applying), 3x2=6 Marks]

- A glass wedge (air film) of 0.01 radian is illuminated by monochromatic light of wavelength 600 nm falls normally on it. Applying the concept of interference at what distance from the edge of wedge, 6th fringe will be observed by the reflected light?
- In Michelson's interferometer experiment, if one of the mirror is moved by a distance of 0.08 mm, 400 fringes cross the field of view. Calculate the wavelength of light using the principles of interference.
- A diffraction grating 3 cm wide produces the second order principal maxima at 30° with light of wavelength 600 nm. What is the total number of lines required on the grating?

Q4 [CO4 (Analyzing), 1x3=3 Marks]

Considering opaque portion in grating lines 'a' is two times of slit width 'b', draw and analyze the intensity pattern formed by three slits and four slits.

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name: Krishna Wadhwa

Enrollment No. 23103060

Jaypee Institute of Information Technology, Noida
T2 Examination, Odd Semester 2023
B. Tech. 1st Year - Semester 1st

Course Title: Physics-I

Course Code: 15B11PH111

Maximum Time: 1 Hour

Maximum Marks: 20

Course outcome: At the completion of the course, students will be able to

CO1	Recall the basic principles of physics related to optics, relativity, radiation, atomic physics and quantum mechanics.
CO2	Illustrate the various physical phenomena with interpretation based on the mathematical expressions involved.
CO3	Apply the concepts/principles to solve the problems related to the wave nature of light, relativity, radiation, atomic physics and quantum mechanics.
CO4	Analyze and examine the solutions of the problems using physical and mathematical concepts involved.

Q1. [CO1 (Remembering) (5 x 1 Mark = 5 Marks)]

State/explain the followings

- Negative and positive uniaxial crystals.
- Double refraction.
- Optical rotation by Fresnel's theory.
- Postulates of special theory of relativity.
- Inverse Lorentz transformation equations.

Q2. [CO2 (Understanding) (3 x 2 Marks = 6 Marks)]

- A light is refracted at 30° through a glass slab. The reflected light is plane polarized having vibrations perpendicular to the plane of incidence in air, find the angle of polarization and refractive index of the glass slab.
- A length of 3 decimeter optically active solution with 10% concentration causes the optical rotation of 15° . How much length of 20% concentration of the same solution will produce 20° rotation?
- The square lamina of side L is measured by an observer at rest. If second observer is moving with velocity ' v ' relative to the first and parallel to one side of the lamina, calculate the area of the lamina as measured by the second observer.

Q3. [CO3 (Applying) (2 x 3 Marks = 6 Marks)]

- Using two Nicole prisms, how would you determine a transparent plate is QWP, HWP or simple glass plate? Discuss it with the help of supporting diagrams.
- Derive the expressions for velocity addition using Lorentz transformation equations. Further determine the relative velocity of a photon and electron (with velocity $0.8c$) moving towards each other. Here c is the speed of light.

Q4. [CO4 (Analyzing) (3 Marks)]

A HWP is introduced between two crossed Polaroids P_1 and P_2 . The optic axis makes an angle of 30° with the pass axis of P_1 . If an unpolarized beam of intensity 50 W/m^2 is incident normally on P_1 , Analyze the intensities I_1 , I_2 and I_3 of light coming out from P_1 , HWP and P_2 , respectively.

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name: ASHNA YADAV

Enrollment No. 23102056

Jaypee Institute of Information Technology, Noida
End Semester Examination, Odd Sem 2023
B. Tech. I Semester

Course Name: Physics-1
 Course Code: 15B11PH111

Max Marks: 35
 Max Time: 02 Hrs

Course Outcomes: After the completion of course, students will be able to

- CO1: Recall the basic principles of physics related to optics, relativity, quantum mechanics, atomic physics.
- CO2: Illustrate the various physical phenomena with interpretation based on the mathematical expressions involved.
- CO3: Apply the concepts/principles to solve the problems related to wave nature of light, relativity, quantum mechanics and atomic physics.
- CO4: Analyze and examine the solution of the problems using physical and mathematical concepts involved.

Note: Answer all questions sequentially and all symbols have their usual meanings.

- Q1a. What is quarter wave plate used in polarization?
 b. What is the shape of interference fringes observed in Young's double slit experiment?
 c. State the physical significance of wave function.
 d. Write down any two applications of tunnel diode.
 e. Can a sound wave be polarized? Justify.

[CO1, Remembering, 5x1 = 5 marks]

- Q2a. Explain non-existence of electron in nucleus by using Heisenberg's uncertainty principle.
 b. Explain ultraviolet catastrophe using appropriate expressions and proper diagram.
 c. Prove that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation.
 d. Derive Rayleigh-Jeans law and Wien's distribution law from Planck's law of black body radiation.
 e. A particle of mass 'm' is trapped in 1D box of width L. Write the wave functions of first three states. Draw their corresponding wave functions and probability densities.

[CO2, Understanding, 5x2 = 10 marks]

- Q3a. Show that group velocity of matter waves associated with a particle is equal to particle velocity, $v_g = v$ and the phase velocity $v_p = c^2/v$. Justify that the phase velocity $v_p > c$ for matter waves.
 b. Find the number of lines in a 2.54 cm wide plane transmission grating, which will just resolve two sodium lines (589 nm and 589.6 nm) in first order.
 c. Calculate the possible orientations of total angular momentum $\vec{J}$ for states of spin $s = 1/2$ and orbital angular momentum number $l = 1$ with respect to the magnetic field along z-axis.
 d. The wave function for a quantum particle of mass 'm' confined to move within $0 \leq x \leq L$ is given by $\psi(x) = N \sin(4\pi x/L)$. Find i) normalization constant of wave function ii) average value of 'x' for particle and iii) calculate the probability of finding the particle in the region from $x=0$ to $x=L/4$.

[CO3, Applying, 4x2.5 = 10 marks]

- Q4a. Derive the expression of wavelength shift in Compton scattering. If the Compton wavelength shift is twice of wavelength of incident photon and photon scattering angle is 60° , find the direction of recoiled electron.
 b. Analyze the state of polarization (give suitable diagram too) of each of the following waves,
 (i) $\vec{E} = 2E_0 \cos(\omega t - kz + \pi/2)\hat{i} + E_0 \sin(\omega t - kz)\hat{j}$
 (ii) $\vec{E} = E_0 \sin(kz - \omega t + \pi/3)\hat{i} + E_0 \sin(kz - \omega t - \pi/6)\hat{j}$
 c. An electron beam is incident on a 1-D barrier of height 6 eV and width 0.2 nm. Find the energy of incident beam to achieve transmission probability of 1%.
 d. An element is placed in 0.5 T magnetic field for suitable excitation. Estimate the Zeeman separation of spectral lines of the element having wavelength 540 nm.

[CO4, Analyzing, 4x2.5 = 10 marks]

Constants: $h = 6.6 \times 10^{-34}$ J-s, $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C, $m_e = 9.1 \times 10^{-31}$ kg.

$$\frac{1}{2m} \frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)} \quad \left(\frac{1}{2}m\right)$$

$z_1 = \infty, z_2 = i, z_3 = 0$
 $w_1 = 0, w_2 = i, w_3 = \infty$

$$\frac{(w-w_1)\left(\frac{w_2}{w_3}-1\right)}{(w_1-w_2)\left(1-\frac{w}{w_3}\right)} = \frac{\left(\frac{z}{z_1}-1\right)(z_2-z_3)}{\left(1-\frac{z_2}{z_1}\right)(z_3-z)} \Rightarrow w = -\frac{1}{z} \quad (1m)$$

$$\frac{2}{2m} \quad u_x = 2(x-y), u_y = -2(x-y), v_x = 2, v_y = 2 \quad (1m)$$

C.R. eq's are not satisfied. $(1m)$

$$\frac{3}{3m} \quad u_x = \frac{2x}{x^2+y^2} + 1, u_y = \frac{2y}{x^2+y^2} + 1 \quad \left. \vphantom{\frac{3}{3m}} \right\} (1m)$$

using C.R. eq's, $u_x = v_y, u_y = -v_x$

$$u_y = -v_x = -\frac{2x}{x^2+y^2} - 1 \Rightarrow u = \tan^{-1}(y/x) - y + g(x)$$

$$\Rightarrow u = -2 \tan^{-1}(y/x) - y + x + C \quad (1m)$$

Harmonicity:

$$\left. \begin{aligned} u_{xx} &= -2x^2 + 2y^2 / (x^2+y^2)^2 \\ u_{yy} &= 2x^2 - 2y^2 / (x^2+y^2)^2 \end{aligned} \right\} \Rightarrow u_{xx} + u_{yy} = 0. \quad (1m)$$

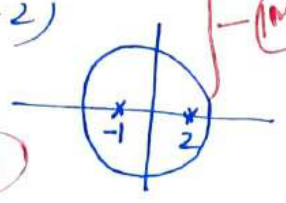
$$\frac{4}{3m} \quad \text{By Cauchy's integral formula for multiply connected domain}$$

$$\int_C f(z) dz = \int_{C_1} \frac{z-1}{(z+1)^2(z-2)} dz + \int_{C_2} \frac{z-1}{(z+1)^2(z-2)} dz \quad \left. \vphantom{\int_C} \right\} (1m)$$

$$= I_1 + I_2$$

$$I_1 = 2\pi i \frac{d}{dz} \left[\frac{z-1}{z-2} \right]_{z=-1} = -\frac{2}{9} \pi i \quad (1m)$$

$$I_2 = 2\pi i \left(\frac{1}{9} \right) \quad \left(\frac{1}{2}m \right)$$

$$\int_C f(z) dz = 0 \quad \left(\frac{1}{2}m \right)$$


$$\frac{5}{3m} \quad f(z) = \frac{z-1}{(z+2)(z+3)} = \frac{-3}{z+2} + \frac{4}{z+3} \quad \left(\frac{1}{2}m \right)$$

(i) $2 < |z| < 3, f(z) = -\frac{3}{z} \left(1 + \frac{z}{2}\right)^{-1} + \frac{4}{3} \left(1 + \frac{z}{3}\right)^{-1}$

$$= -\frac{3}{z} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \dots\right) + \frac{4}{3} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \dots\right) \quad (1m)$$

(ii) $|z+1| < 1$

$$f(z) = -3 \left[1 + (z+1)\right]^{-1} + 2 \left[1 + \frac{z+1}{2}\right]^{-1}$$

$$= -3 \left[1 - (z+1) + (z+1)^2 - \dots\right] + 2 \left[1 - \frac{z+1}{2} + \left(\frac{z+1}{2}\right)^2 - \dots\right]$$

Taylor's series $\left(\frac{1}{2}m \right)$

6. $\frac{6}{3M}$ $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta} = \oint_C \frac{1}{5-4(\frac{z-1}{2iz})} \frac{dz}{iz}$ (1M)
 $= -\oint_C \frac{dz}{2z^2 - 5iz - 2}$, $z = 2i, \frac{i}{2}$ (1M)
inside
 $\text{Res}[z = \frac{i}{2}] = -\frac{1}{3i}$
 $\therefore I = -2\pi i (-\frac{1}{3i}) = \frac{2\pi}{3}$ (1M)

7a. $\frac{7a}{2M}$ $\sum U_n(x)$ is uniformly convergent if a convergent series $\sum M_n$ of +ve real no $|U_n(x)| \leq M_n$ (2M)

b. $\frac{b}{3M}$ $U_n = \frac{(x-1)^n}{2^n}$, $U_{n+1} = \frac{(x-1)^{n+1}}{2^{n+1}}$, $\lim_{n \rightarrow \infty} \left| \frac{U_n}{U_{n+1}} \right| = \frac{2}{|x-1|}$ (1M)

(i) if $\frac{2}{|x-1|} > 1 \Rightarrow |x-1| < 2$ convergent } (1M)

(ii) if $|x-1| > 2$ divergent

at $|x-1| = 2 \Rightarrow x-1 = \pm 2$

$x-1 = 2 \Rightarrow x = 3$

$\sum 1 = \text{divergent}$

$\sum (-1)^n = \text{oscillatory } (-1, 3)$ (1M)

$x-1 = -2 \Rightarrow x = -1$

$x = 1$

8. $\frac{8}{5M}$ $y = uv$; $u = \frac{1}{x}$; $I = 1$ (1M)
 eqn $v'' + v = \sin 2x$ (1M) CF = $C_1 \cos x + C_2 \sin x$ (1M)
 $v = C_1 \cos x + C_2 \sin x - \frac{1}{3} \sin 2x$ (1M) $y = uv$ (1M)

9. (a) $\frac{9}{3M}$ $(1-2xh+h^2)^{1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$ (1M)

$(1+2xh+h^2)^{1/2} = \sum_{n=0}^{\infty} h^n P_n(-x)$ (2) (1M)

$(1+2xh+h^2)^{1/2} = \sum_{n=0}^{\infty} (-h)^n P_n(x)$ (3) (1M)

from (2) & (3) $P_n(-x) = (-1)^n P_n(x)$ (1M)

(b) $\frac{b}{2M}$ Singular points are $x=0, 1$ (1M)

$x=0$ regular singular (1M) | irregular singular (1M)

10. $\frac{10}{4M}$ $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial u}{\partial t}$ $u(x,t) = (C_1 \cos px + C_2 \sin px) e^{-\frac{p^2 c^2 t}{2}}$ (1M)

$C_1 = 0$, $p = \frac{n\pi}{l}$ (1M)

$u(x,0) = \sin \frac{4\pi x}{l} \cos \frac{\pi x}{l}$
 $\sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} = \frac{1}{2} \left(\sin \frac{3\pi x}{l} + \sin \frac{\pi x}{l} \right)$ (1M)

$b_1 = \frac{1}{2}$, $b_3 = \frac{1}{2}$

$u(x,t) = \frac{1}{2} \left[\sin \frac{\pi x}{l} e^{-\frac{\pi^2 c^2 t}{l^2}} + \sin \frac{3\pi x}{l} e^{-\frac{9\pi^2 c^2 t}{l^2}} \right]$ (1M)